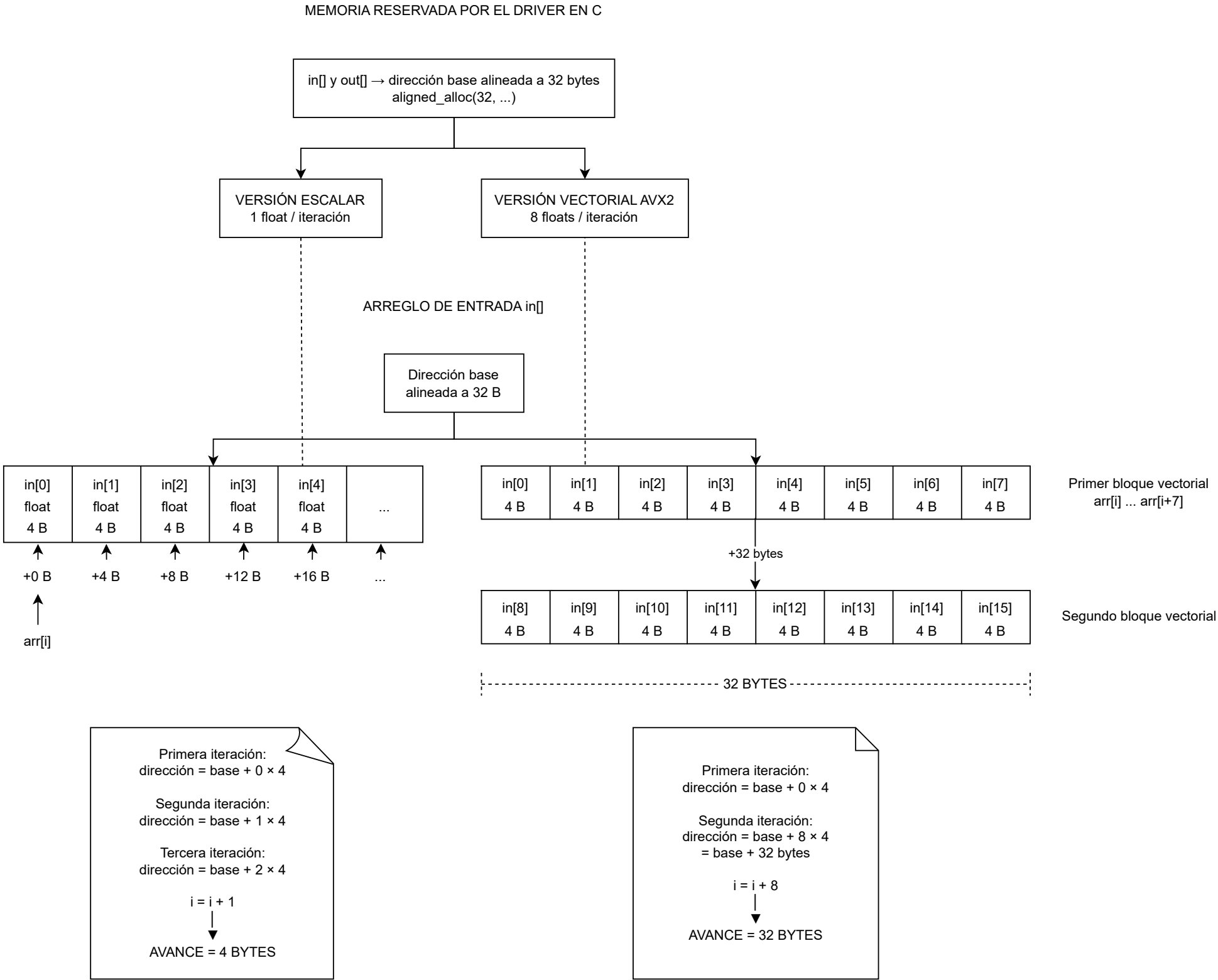
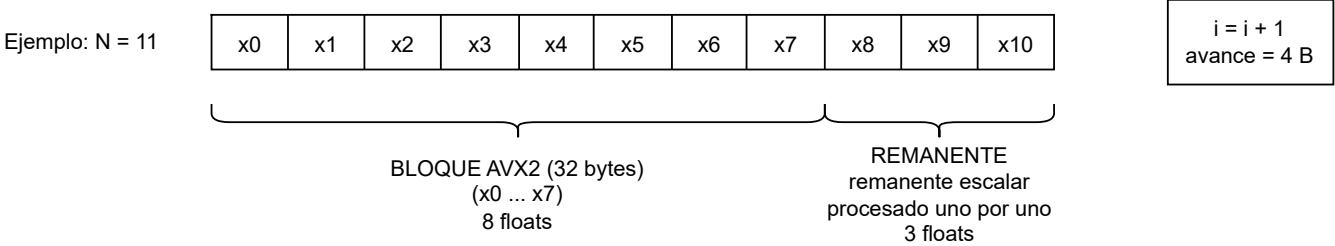


DIAGRAMA 4 — MAPA DE MEMORIA Y ALINEACIÓN
ESCALAR VS. VECTORIAL AVX2



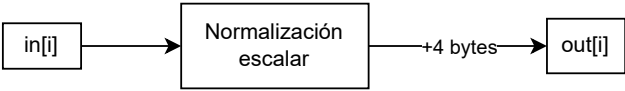
REMANENTE VECTORIAL



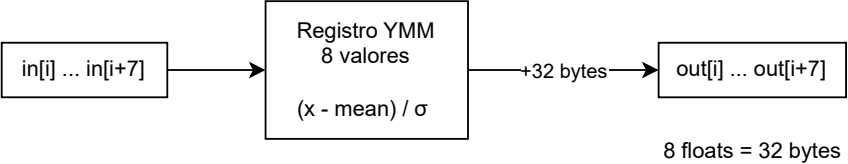
ARREGLO DE SALIDA out[]

$$\text{out}[i] = \frac{\text{in}[i] - \text{mean}}{\text{stddev}}$$

FLUJO ESCALAR



FLUJO VECTORIAL AVX2



RESUMEN DE ACCESO A MEMORIA

	ESCALAR	VECTORIAL AVX2
Tamaño float	4 bytes	4 bytes
Datos/iteración	1 float	8 floats
Avance índice	i = i + 1	i = i + 8
Avance memoria	4 bytes	32 bytes
Registro datos	XMM (parte baja)	YMM (256 bits)
Remanente	No aplica	Escalar, 1 a la vez
Base in[]	Alineada a 32 B	Alineada a 32 B
Base out[]	Alineada a 32 B	Alineada a 32 B